

SmartBin-1

Prototype Costing Report

Automated IoT Waste-Level Monitoring & Alert System · Prototype V1.0

REPORT DATE

01 Oct 2026

CURRENCY

Indian Rupee (₹)

PREPARED BY

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STATUS

Final · prototype build

₹870.45

TOTAL EXPENDITURE

13

LINE ITEMS BOUGHT

37

TOTAL PIECES PURCHASED

60.8%

SHARE: ESP32 BOARD

01

ITEMISED BILL

#	ITEM & SPECIFICATION	CATEGORY	QTY	RATE (₹)	AMOUNT (₹)
01	Solderless breadboard 830-pt, dual rail	Prototyping	2	70.00	140.00
02	Carbon film resistor 220 Ω 1/4 W · LED current limit	Passives	3	1.00	3.00
03	Carbon film resistor 1 kΩ 1/4 W 5% · ECHO divider	Passives	3	1.00	3.00
04	LED green, diffused 3 mm · OK state	Sensing & Alerts	1	1.60	1.60
05	LED yellow, diffused 3 mm · HALF FULL	Sensing & Alerts	1	1.60	1.60
06	LED red, round 5 mm · FULL state	Sensing & Alerts	1	2.00	2.00
07	Active buzzer alarm module DC 3.3-5 V · ≥80% alarm	Sensing & Alerts	1	29.00	29.00
08	HC-SR04 ultrasonic sensor Distance ranging	Sensing & Alerts	1	75.00	75.00
09	Jumper wire, female-female 2.54 mm · 20 cm	Wiring	7	1.25	8.75
10	Jumper wire, male-male 2.54 mm · 20 cm	Wiring	14	1.25	17.50
11	ESP32 DevKit V1 + cable ESP-WROOM-32 · controller	Controller	1	529.00	529.00
12	Double-sided tape Sensor / board mounting	Enclosure	1	10.00	10.00
13	Plastic dustbin container Calibrated 14.4 cm bin	Enclosure	1	50.00	50.00

GRAND TOTAL

37 pieces · 13 line items

₹ 870.45

02

COST BREAKDOWN BY CATEGORY

Controller		₹529.00	60.8%
Prototyping		₹140.00	16.1%
Sensing & Alerts		₹109.20	12.5%
Enclosure		₹60.00	6.9%
Wiring		₹26.25	3.0%
Passives		₹6.00	0.7%

COST INSIGHTS

₹800.45

single-node cost (one breadboard counted; 2nd is spare)

₹109.20

entire sensing + alert stack (sensor, buzzer, 3 LEDs)

₹201.45

everything except ESP32 and breadboards

NOTES

Amounts are the line totals paid; the rate column is derived as amount ÷ quantity (e.g. jumpers ₹1.25 each, resistors ₹1.00 each).

Two breadboards were bought; the project report's BOM lists one — the extra ₹70.00 is spare/prototyping stock.

LED sizes bought: 3 mm green & yellow, 5 mm red (report BOM lists 5 mm). Prices are as paid; taxes/shipping not separately itemised.